

The objective of this machine learning analysis is to predict developer job satisfaction (*JobSat*) using the Stack Overflow Developer Survey. Because job satisfaction is measured as an ordinal outcome rather than as a continuous variable, the analysis is considered an ordinal classification task. Beyond building a predictive model, the analysis also aims to identify which survey attributes are most strongly associated with job satisfaction and to evaluate how reliably these patterns generalize from the training data to unseen test data.

To achieve a reliable evaluation, the dataset was first divided into training and test sets using an 80/20 split. This design helps prevent data leakage from the test set into the model-development process and ensures that the final performance metrics reflect genuine out-of-sample prediction. All preprocessing decisions were learned using the training data and then transferred to the test set. However, within cross-validation, feature selection and scaling were applied to the prepared training feature set rather than being refit inside each fold, so the cross-validation estimates should be interpreted as approximate.

Missing values were handled separately for categorical and numerical variables. For categorical variables, missing values were replaced with an “Unknown” category because non-response may be informative and may reflect inapplicability, uncertainty, or intentionally skipped questions. For numerical variables, columns with more than 70% missing values were removed (*AgentUsesGeneral*, *AIAgentKnowledge*, *AIAgentOrchestration*, *AIAgentObserveSecure*, and *AIAgentExternal*), while the remaining numerical features were imputed using the training-set median. This threshold was chosen because once most values in a variable are missing, imputation is more likely to introduce substantial artificial bias as noise than useful information. Median imputation was selected because it is robust to outliers and is well-suited to skewed survey-based variables such as compensation and experience. Considering the reason for these missing values, the very high missingness variables were AI-Agent-related. They likely reflect question inapplicability or selective non-response, since not all respondents used AI agents or encountered those workflows. Keeping these variables through aggressive imputation would likely add more noise than a meaningful signal.

After missing-value handling, the categorical survey variables were encoded. Single-response categorical variables were transformed using one-hot encoding because most of these features, such as country, employment type, and organizational role, are nominal and do not have a natural ordering. One-hot encoding avoids introducing artificial rank or distance relationships between categories and allows the model to treat each category independently. For multi-response variables, an exploratory mutual-information analysis was first used to examine which multi-response columns were most relevant to *JobSat*. This step was used for interpretation rather than for filtering. The multi-response columns, such as *LanguageChoice*, *DatabaseChoice*, and *PlatformChoice*, were then converted using multi-hot encoding so that each possible response option became its own binary indicator. This method consciously avoided the chance of dealing with a complex mix of answers as a single category; instead, it made sure that every single individual answer that the respondent chose had the opportunity to contribute its own predictive signal.

Because the majority of predictors were represented as non-negative binary features after encoding, the chi-square test could serve as a highly appropriate and computationally efficient method to evaluate the dependency between the vast majority of our feature space and *JobSat*. Continuous numerical variables were excluded from this ranking because large-scale numerical features could distort chi-square scores. The chi-square ranking illustrated that several feature groups, including career-transition intention, employment status, workplace context, and AI-related perceptions, were associated with job satisfaction (Fig. 1). These patterns suggest that job satisfaction is shaped not only by technical background, but also by broader professional circumstances such as whether respondents are considering career changes, how secure they feel about the role of AI, and the organizational environment in which they work. Based on these patterns, three new features were engineered (*ExperienceGap*, *CompPerYearExp*, and *ConsideringCareerChange*). These variables were designed to capture broader relationships that might not be explicitly represented by the encoded survey features alone. Specifically, *ExperienceGap* represents the gap between total coding experience and work experience; *CompPerYearExp* represents compensation per year of work experience, and *ConsideringCareerChange* represents whether the respondent was considering a career change.

Given the high dimensionality and sparsity of the encoded survey data, Elastic Net regularization was applied for feature selection. Compared with an unregularized regression approach or PCA, Elastic Net can handle a large number of correlated features and control model complexity while retaining informative variables by shrinking coefficients and removing weak predictors. After the regularization, the feature space was reduced to 199 predictors (Fig. 2). Notably, among the three newly engineered features, only *ExperienceGap* was dropped, whereas *ConsideringCareerChange* was illustrated as a highly leading predictor with a coefficient of -0.337, which suggests that a developer actively considering a career change serves as one of the strongest quantifiable signals of job dissatisfaction. The same engineered features and selected feature set were then applied to the test data so that the final training and test feature matrices had matching dimensions for the later modeling stage.

After preprocessing, an ordinal logistic regression algorithm was implemented. This model was chosen because the target variable represents ordered satisfaction levels rather than an unordered categorical outcome. Before training, the selected features were standardized using the training-set mean and standard deviation, and the same transformation was then applied to the test set. Standardization was useful because the final feature set contained both continuous variables and binary encoded indicators. Without standardization, variables on larger numerical scales could dominate the optimization process and make coefficient estimation less stable.

After the baseline training using the default hyperparameter sets, the model was evaluated with 10-fold cross-validation on the training set. It achieved a mean accuracy of 0.2718 with a variance of 0.0000 across folds, which indicates that fold-to-fold differences were extremely small and that the baseline model behaved consistently across different training subsets. However, despite this stability, the overall predictive performance remained limited. Therefore, the next step was to examine a hyperparameter that directly controls model complexity and could potentially improve the balance between underfitting and overfitting.

In this model, the regularization parameter C was chosen for that purpose because it directly affects the bias-variance trade-off. Smaller values of C strengthen regularization, which constrains the model, increases bias, and reduces variance. Larger values weaken regularization, which allows a more flexible fit that may reduce bias but increase variance. The parameter C was varied over the values 0.01, 0.05, 0.1, 0.5, 1, 2, 5, and 10, while the remaining settings were held fixed. For each value of C , the model was evaluated using 10-fold cross-validation, and both the mean accuracy (Fig. 3) and the variance (Fig. 4) across folds were recorded. The results showed that cross-validation accuracy changed only slightly across the tested values of C , with all mean accuracies remaining close to 0.27 and all variances remaining very small. This suggests that the model was relatively insensitive to C over the tested range and that changing the regularization strength did not substantially alter the practical balance between bias and variance. Among the tested values, $C = 0.01$ produced the highest mean accuracy (0.2722), although the difference from the other values was very small. Therefore, $C = 0.01$ was selected as the best value from the bias-variance comparison.

To further complement this cross-validation-based analysis, the provided *bias_variance_decomp* helper function was also applied across the same range of C values (Fig. 5 and Fig. 6). The decomposition results were consistent with the cross-validation findings. Within the tested values, $C=0.01$ achieved both the lowest estimated bias (3.6251) and the lowest estimated variance (0.3584). As C increased, the estimated variance generally increased, while the estimated bias only slightly changed across the tested range. This suggests that weaker regularization did not provide a meaningful reduction in bias, but did introduce greater prediction instability. Therefore, $C=0.01$ remained the most favorable value for the bias-variance trade-off.

Because varying C itself only produced a limited improvement, a broader hyperparameter tuning step was conducted using grid search. This allowed the model to be evaluated across combinations of hyperparameter values and provided a more systematic way to identify the final best-performing model. In terms of performance metrics, accuracy was a misleading metric for this dataset because it only measures the overall proportion of correct predictions and can be influenced by the most frequent satisfaction levels. As a result, a model may achieve a moderate accuracy score by predicting the majority classes more often while still performing poorly on the less frequent classes. Therefore, the Macro F1-score was used as an alternative metric because it gives equal weight to each class by averaging the F1-scores across all classes regardless of sample size. This makes it more suitable for evaluating whether model performance is reasonable across the full range of job-satisfaction levels rather than only on the dominant classes.

The grid search was conducted over C and max_iter . The parameter C was kept because it is the main regularization parameter of the model and had already been analyzed in relation to the bias-variance trade-off. The second parameter, max_iter , controls the maximum number of optimization iterations and was included to ensure stable convergence for the high-dimensional encoded dataset. To make hyperparameter tuning more appropriate for the imbalanced JobSat classes, StratifiedKFold was also included to ensure that each fold contained a similar class distribution and made the Macro F1-based evaluation more reliable across all satisfaction levels.

The best-performing model had the optimal hyperparameters $C = 5$ and $\text{max_iter} = 500$; it also achieved a best cross-validation Macro F1-score of 0.1315. This result suggests that weaker regularization allowed the model to capture more complex patterns in the training data without being overly constrained. However, the relatively low Macro F1-score also emphasizes the issues of severe class imbalance in the dataset. In particular, the model may struggle to predict the most underrepresented satisfaction levels, which lowers the overall macro-average score. In terms of the improvement, it could potentially come from strategies such as using balanced class weights during training or exploring more flexible nonlinear models, such as tree-based approaches, that may be better able to capture the subjective and complex factors of job satisfaction.

Feature importance for the final tuned ordinal logistic regression model was summarized using the mean absolute coefficient value across the internal binary logistic regression submodels (Fig. 7). The plot showed that *CompPerYearExp* was the most influential engineered feature, followed by *ConsideringCareerChange*, several *NewRole* indicators, and a number of country-specific dummy variables. *Employment*-related, *AI*-related, and *workplace-context* features also remained as leading features. Compared with the earlier exploratory ranking, the final model showed the same importance of *career-transition* intention and *employment context*, while the *compensation-based engineered information* was emphasized after regularized feature selection and model fitting.

After the best model was identified, the final tuned model was evaluated on the held-out test set. The model achieved a training accuracy of 0.2999 and a test accuracy of 0.2787, while the training Macro F1-score was 0.1774 and the test Macro F1-score was 0.1426. These results indicate that the model performed somewhat better on the training data than on unseen test data, but that its overall predictive performance remained modest. The difference between training and test results suggests some overfitting patterns, since model performance was consistently higher on the training set than on the test set (Fig. 8). In particular, the gap between the training and test Macro F1-scores indicates that the model generalized less effectively across all satisfaction classes than it did on the training data. Together, the final model was not severely overfit, but it still struggled to distinguish the full range of JobSat categories effectively.

To further compare the true and predicted label distributions, the class frequencies of the model predictions were plotted against the observed JobSat frequencies for both the training and test sets (Fig. 9). These plots showed that the predicted distribution roughly followed the broad shape of the true distribution, but the model concentrated much more heavily on the most common satisfaction levels, such as class 8 and class 7, and struggled to recover the rarer categories. This again suggests that the ordinal logistic regression model captured some of the main structure in the data, but did not fully distinguish the less frequent satisfaction levels.

One key finding from the final model was that compensation relative to work experience (*CompPerYearExp*) and career-change intention (*ConsideringCareerChange* and *NewRole*-related variables) remained among the strongest predictors of job satisfaction (Fig. 10). This suggests that developer job satisfaction is shaped not only by technical background, but also by broader professional conditions such as whether respondents feel

appropriately compensated for their experience and whether they are actively considering a career transition.

In conclusion, this analysis developed an ordinal machine learning pipeline to predict developer job satisfaction from the Stack Overflow Developer Survey. Although the final ordinal logistic regression model showed stable behavior across cross-validation folds, its predictive performance on unseen data remained modest. The analysis consistently identified career-transition intention, compensation relative to work experience, employment context, and AI-related perceptions as important factors associated with JobSat across exploratory analysis, feature selection, and the final tuned model. Overall, while the model was not highly accurate across all satisfaction levels, it still provided useful insight into the main factors related to developer job satisfaction.

Appendix A

Figure 1. Top 20 Categorical Features by Chi-square Score

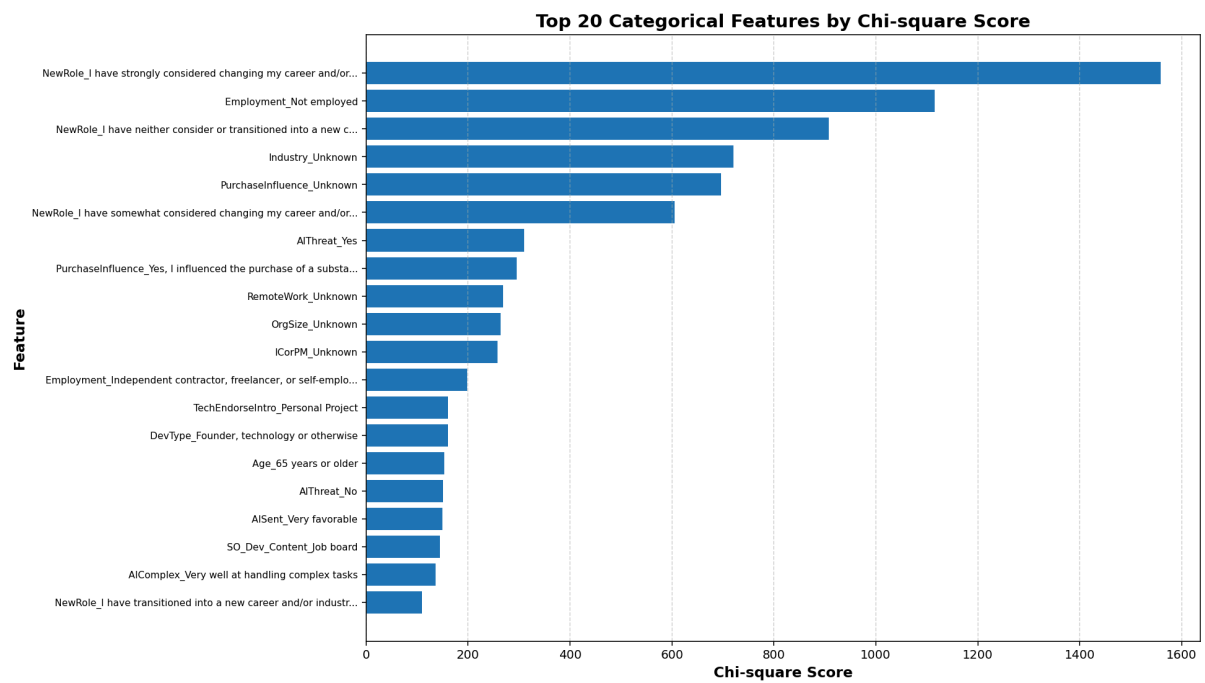


Figure 2. Top 30 Features by ElasticNet Coefficient Magnitude

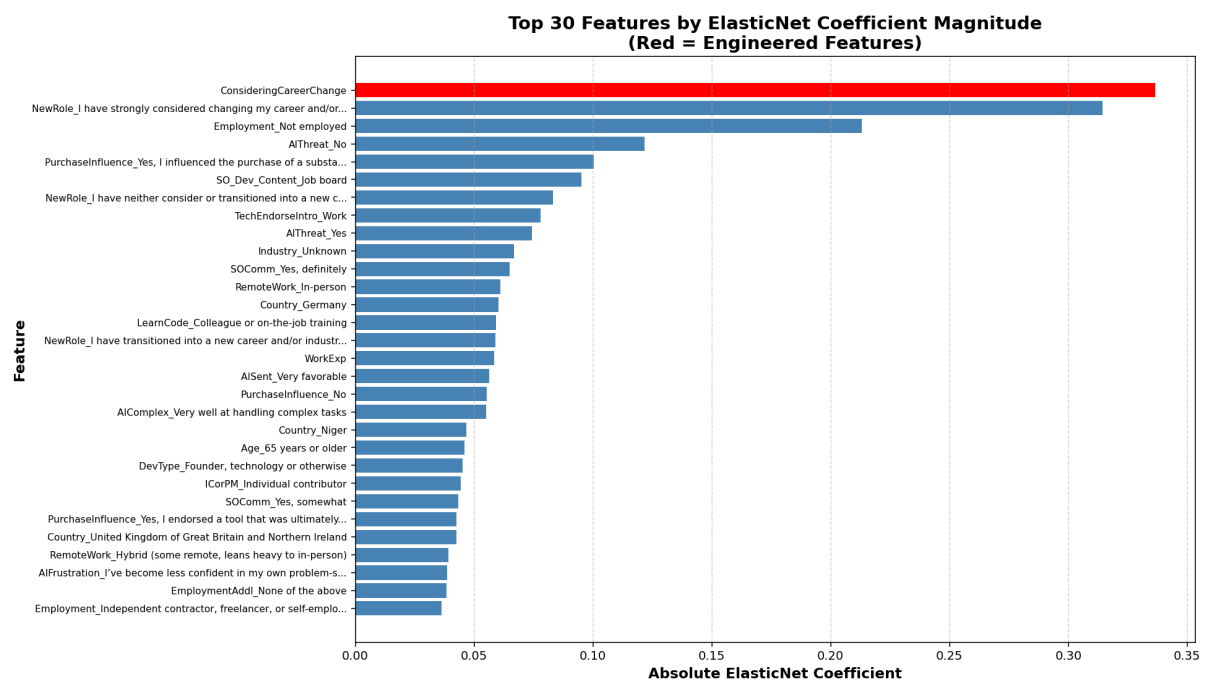


Figure 3. Bias-Variance Trade-off: Effect of C on Mean CV Accuracy

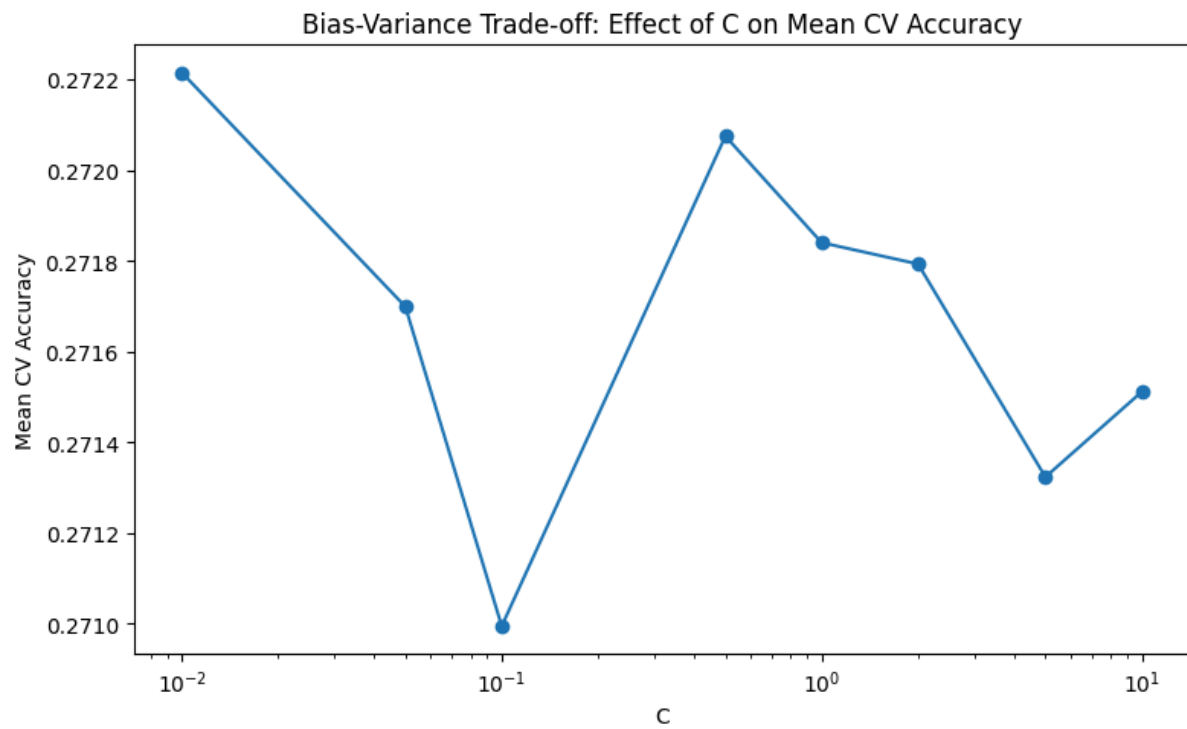


Figure 4. Bias-Variance Trade-off: Effect of C on CV Accuracy Variance

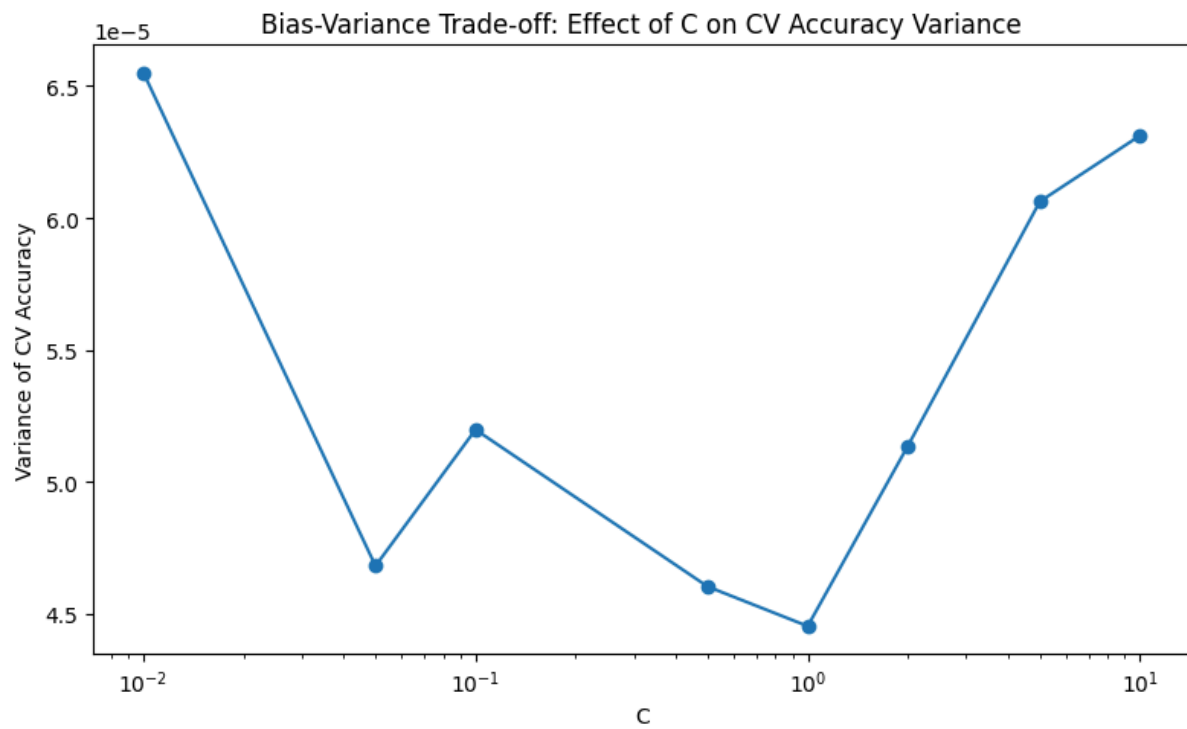


Figure 5. Bias-Variance Decomposition: Effect of C on Estimated Bias

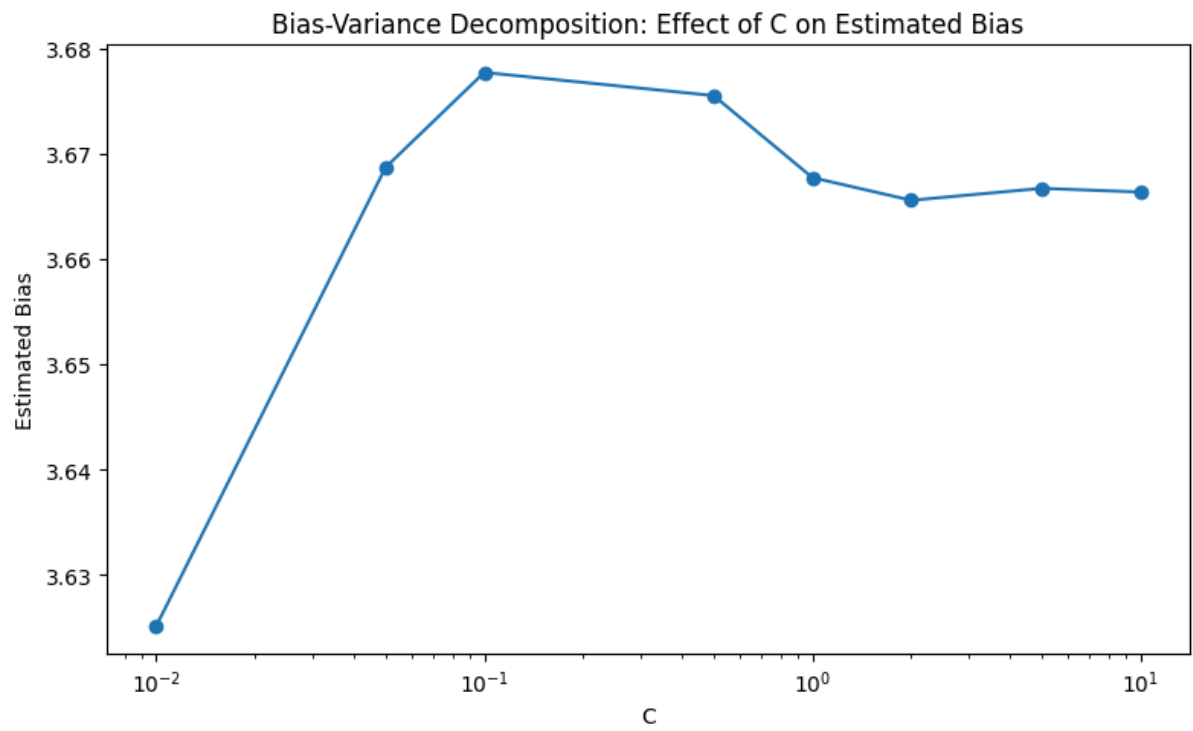


Figure 6. Bias-Variance Decomposition: Effect of C on Estimated Variance

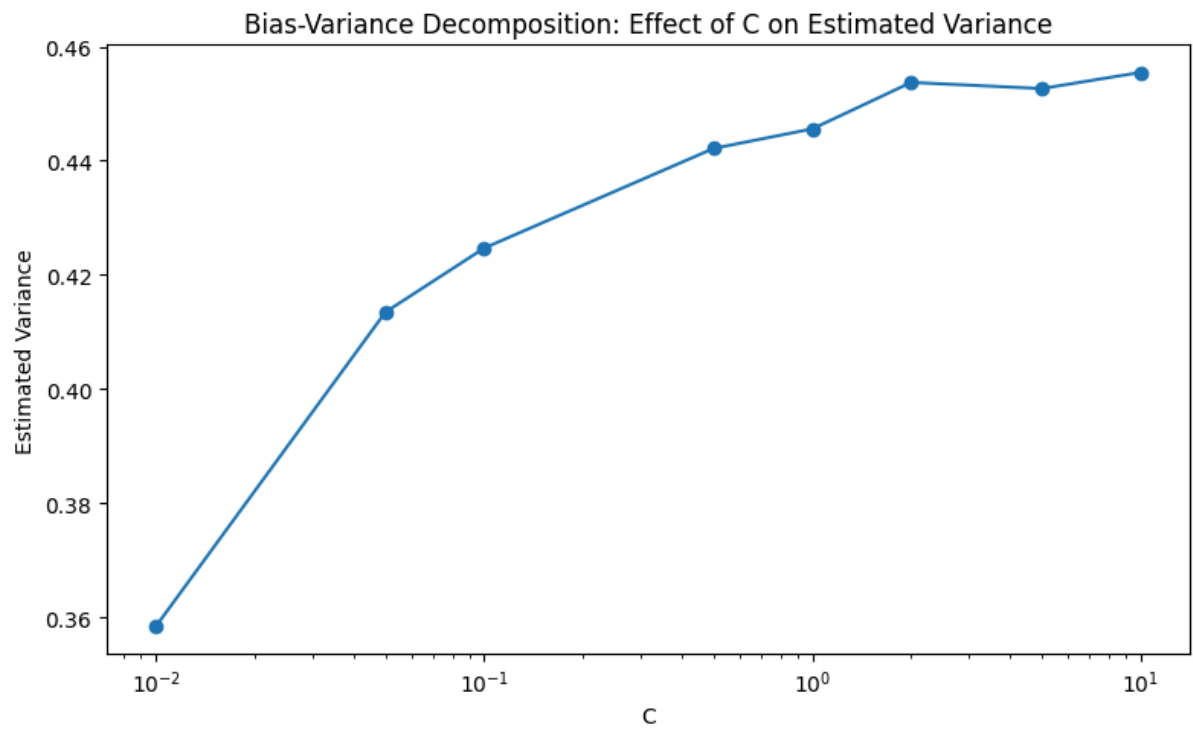


Figure 7. Top 30 Features of the Final Tuned Ordinal Logistic Regression Model

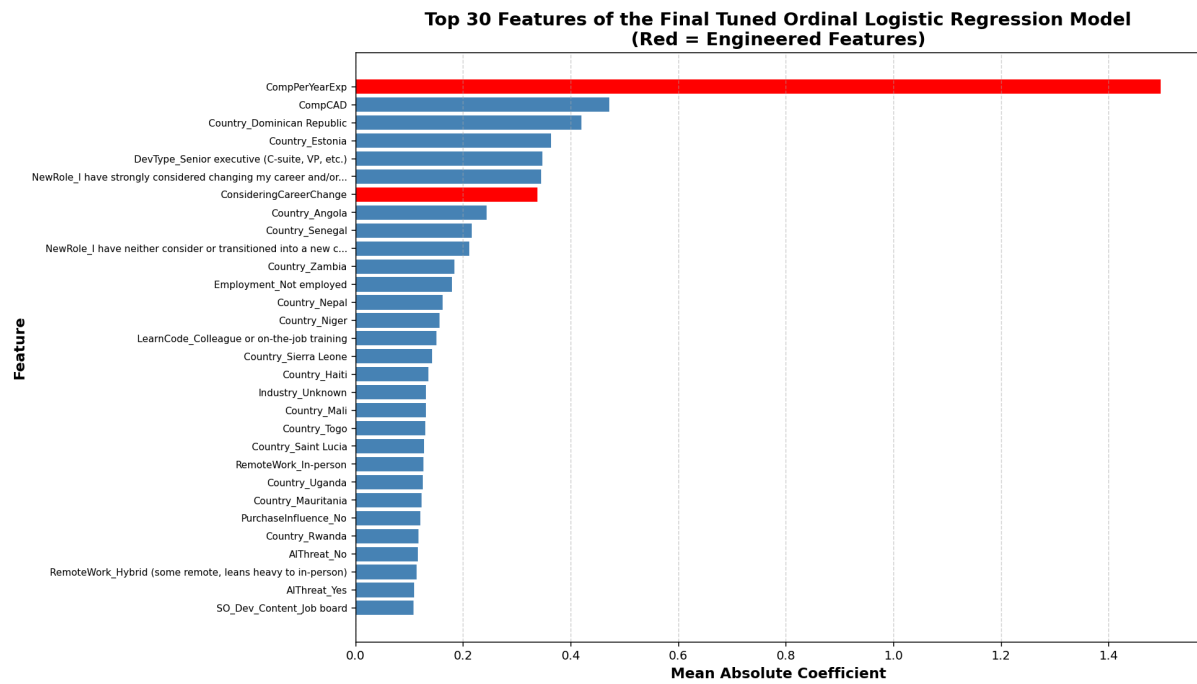


Figure 8. Train vs Test Performance of the Final Model (Accuracy and Macro F1)

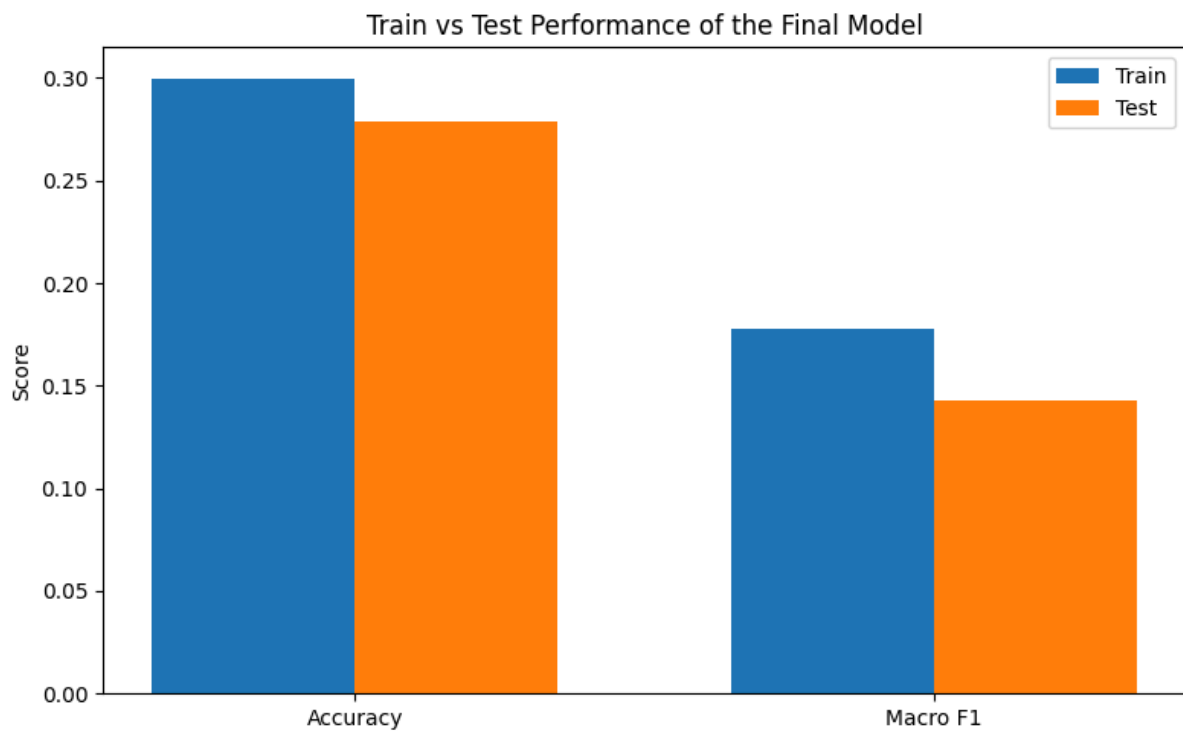


Figure 9. True and Predicted Label Distributions (Train and Test set)

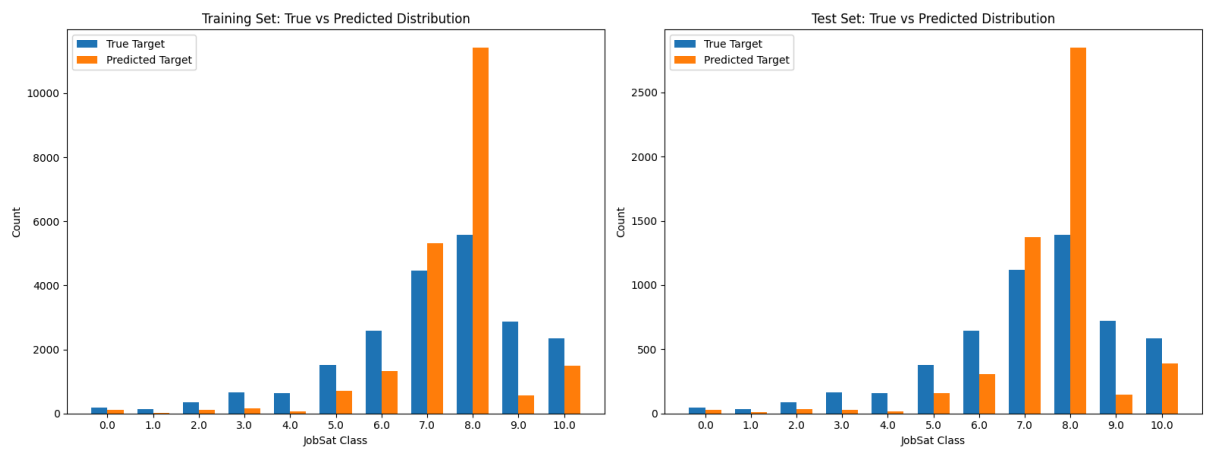
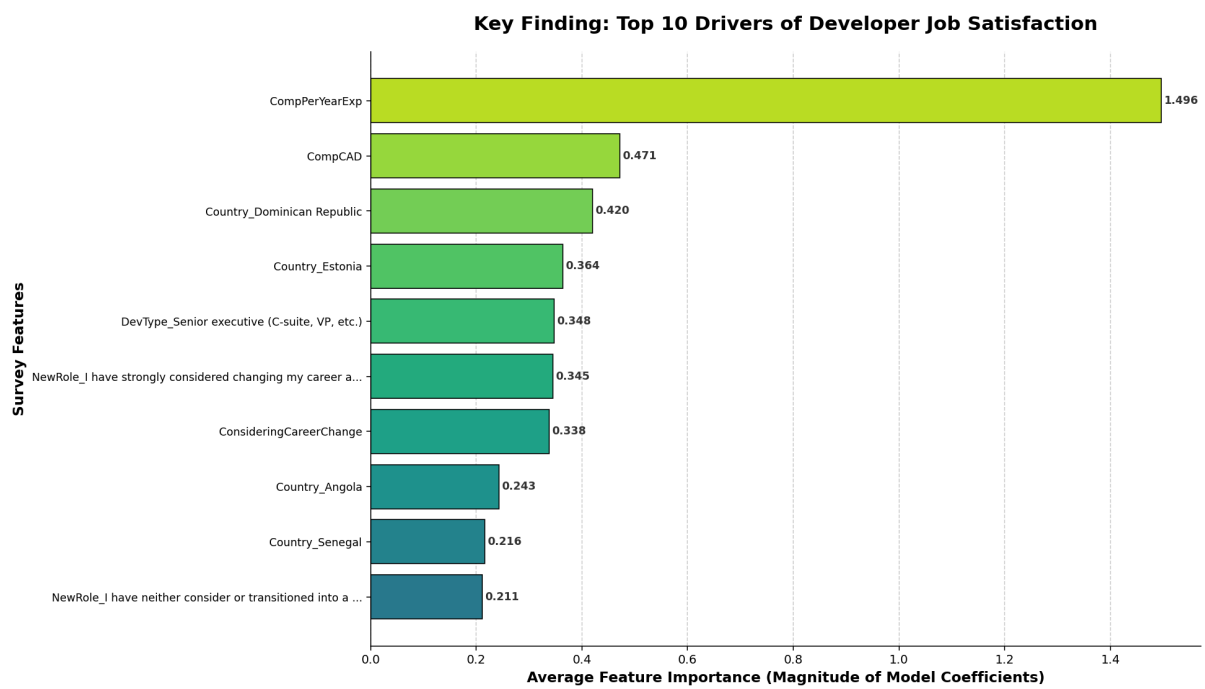


Figure 10. Top 10 Drivers of Developer Job Satisfaction



Statement of LLM Usage

The LLM's contribution was restricted to assisting in the interpretation, wording, and presentation of this report. Some examples of its use include determining whether Elastic Net was better suited than PCA for feature selection, determining whether mutual information could be used as an exploratory analysis for multi-response variables, and examining my statements of Macro F1-score and accuracy's pros and cons. The code, results, figures, and conclusions presented in this report were solely created and validated by me.

A Specific Example of LLM-Assisted Interpretation

A specific application of the LLM in this assignment was in helping me evaluate methodological explanations rather than make the final decisions for me. For example, I used the LLM to ask if Elastic Net would be more appropriate than using PCA after the application of one-hot and multi-hot encoding, given the sparsity and high-dimensionality of the feature space. I also used the LLM to ask if mutual information could be used first as a pre-processing step to evaluate which of the multi-response survey columns were most relevant to JobSat, rather than treating the combined responses as single categories right away.